$i3 = [0, 0, 1]^T$ $i1 = [1, 0, 0]^T$

$$T_{k-1}^{k}(\theta_k, d_k, a_k, \alpha_k) = Rot(\theta_k, 3) Tran(d_k i^3) Tran(a_k i^1) \underline{Rot(\alpha_k, 1)}$$

$$Rot(\theta_k, 3) = \begin{pmatrix} \cos\theta_k & -\sin\theta_k & 0 & 0 \\ \sin\theta_k & \cos\theta_k & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Tran(d_k i^3) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_k \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Tran(a_k i^1) = \begin{pmatrix} 1 & 0 & 0 & a_k \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\underline{Rot(\alpha_k, 1)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_k & -\sin\alpha_k & 0 \\ 0 & \sin\alpha_k & \cos\alpha_k & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_{k-1}^{k}(\theta_k, d_k, a_k, \alpha_k) = Rot(\theta_k, 3)Tran(d_k i^3)Tran(a_k i^1)Rot(\alpha_k, 1)$$

$$\begin{pmatrix} \cos\theta_k & -\sin\theta_k & 0 & 0 \\ \sin\theta_k & \cos\theta_k & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_k \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & a_k \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_k & -\sin\alpha_k & 0 \\ 0 & \sin\alpha_k & \cos\alpha_k & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} C\theta_k & -C\alpha_k S\theta_k & S\alpha_k S\theta_k & a_k C\theta_k \\ S\theta_k & C\alpha_k C\theta_k & -S\alpha_k C\theta_k & a_k S\theta_k \\ 0 & S\alpha_k & C\alpha_k & d_k \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Example 2.11:

Axis	θ	d	a	α
1	θ_1	d_1	0	$-\pi/2$

C_1 S_1

For $k = 1$

$$T_0^1 = \begin{pmatrix} C\theta_1 & -C\alpha_1 S\theta_1 & S\alpha_1 S\theta_1 & a_1 C\theta_1 \\ S\theta_1 & C\alpha_1 C\theta_1 & -S\alpha_1 C\theta_1 & a_1 S\theta_1 \\ 0 & S\alpha_1 & C\alpha_1 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow = \begin{pmatrix} C\theta_1 & 0 & -S\theta_1 & 0 \\ S\theta_1 & 0 & C\theta_1 & 0_1 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\boxed{\begin{aligned} &a_1 = 0, \\ &\alpha_1 = -90 \;\Rightarrow C\alpha_1 = 0 \quad S\alpha_1 = -1 \end{aligned}}$$